

Challenge: Ramnit Lab

Platform: SOC Analyst Tier 1

Category: Endpoint Forensics

Tactics: Initial Access, Execution

Tools: Volatility 3, VirusTotal

VIEWS LAB: [link](#)

Kịch bản: Analyze a memory dump using Volatility to identify a malicious process, extract network IOCs, file hash, and compilation timestamp, correlating with external threat intelligence.

Q1

What is the name of the process responsible for the suspicious activity?

1. Ngoại lệ là một quy trình trông hơi đáng ngờ: một **ChromeSetup.exe** đang chạy khi việc ghi hình được thực hiện. Vì hệ thống phát hiện xâm nhập (IDS) đã cảnh báo về nghi ngờ phần mềm độc hại được thực thi, hãy bắt đầu bằng việc điều tra quá trình này và ghi chú lại **ID tiến trình** (PID), chúng ta sẽ cần nó cho câu hỏi tiếp theo.

MINGW64/C:/Users/sotha/Downloads/159-Ramnit/temp_extract_dir											
2996	660	svchost.exe	0xca82b3d3da2c0	15	-	0	False	2024-02-01 19:48:25.000000 UTC	N/A	Disabled	
3040	660	svchost.exe	0xca82b3d3f5080	6	-	0	False	2024-02-01 19:48:25.000000 UTC	N/A	Disabled	
2872	660	svchost.exe	0xca82b3e68240	7	-	0	False	2024-02-01 19:48:25.000000 UTC	N/A	Disabled	
3116	1144	taskhostw.exe	0xca82b3e6d300	8	-	1	False	2024-02-01 19:48:25.000000 UTC	N/A	Disabled	
3128	660	svchost.exe	0xca82b3e6d080	7	-	0	False	2024-02-01 19:48:25.000000 UTC	N/A	Disabled	
3160	660	svchost.exe	0xca82b3e9f2c0	2	-	0	False	2024-02-01 19:48:25.000000 UTC	N/A	Disabled	
3208	660	svchost.exe	0xca82b3e6d280	3	-	0	False	2024-02-01 19:48:25.000000 UTC	N/A	Disabled	
3248	660	VGAuthService.exe	0xca82b3e6d300	3	-	0	False	2024-02-01 19:48:25.000000 UTC	N/A	Disabled	
3256	660	vmtoolsd.exe	0xca82b3e6d0c0	5	-	0	False	2024-02-01 19:48:25.000000 UTC	N/A	Disabled	
3288	660	wins.exe	0xca82b3e9f240	3	-	0	False	2024-02-01 19:48:25.000000 UTC	N/A	Disabled	
3296	660	vmtoolsd.exe	0xca82b3e9f280	12	-	0	False	2024-02-01 19:48:25.000000 UTC	N/A	Disabled	
3356	660	svchost.exe	0xca82b3e9fa240	8	-	0	False	2024-02-01 19:48:25.000000 UTC	N/A	Disabled	
3420	660	svchost.exe	0xca82b3f8f2c0	4	-	0	False	2024-02-01 19:48:25.000000 UTC	N/A	Disabled	
3444	3256	vmtoolsd.exe	0xca82b3f8c200	6	-	1	False	2024-02-01 19:48:25.000000 UTC	N/A	Disabled	
3692	660	svchost.exe	0xca82b7061280	6	-	0	False	2024-02-01 19:48:25.000000 UTC	N/A	Disabled	
3852	660	svchost.exe	0xca82b1c90240	13	-	0	False	2024-02-01 19:48:25.000000 UTC	N/A	Disabled	
3936	3692	ctfmon.exe	0xca82b7107240	11	-	1	False	2024-02-01 19:48:25.000000 UTC	N/A	Disabled	
3104	660	svchost.exe	0xca82b7229080	12	-	0	False	2024-02-01 19:48:26.000000 UTC	N/A	Disabled	
4276	660	dllhost.exe	0xca82b7320280	15	-	0	False	2024-02-01 19:48:26.000000 UTC	N/A	Disabled	
4416	660	svchost.exe	0xca82b7408240	34	-	0	False	2024-02-01 19:48:26.000000 UTC	N/A	Disabled	
4492	808	wmiprvse.exe	0xca82b7424280	13	-	0	False	2024-02-01 19:48:26.000000 UTC	N/A	Disabled	
4508	624	userinit.exe	0xca82b7426340	0	-	1	False	2024-02-01 19:48:26.000000 UTC	2024-02-01 19:48:52.000000 UTC	Disabled	
4568	4508	explorer.exe	0xca82b7440340	35	-	1	False	2024-02-01 19:48:26.000000 UTC	N/A	Disabled	
4796	660	svchost.exe	0xca82b758d300	7	-	0	False	2024-02-01 19:48:26.000000 UTC	N/A	Disabled	
4836	660	msdtc.exe	0xca82b75a7280	12	-	0	False	2024-02-01 19:48:26.000000 UTC	N/A	Disabled	
3152	1144	taskhostw.exe	0xca82b7681280	7	-	1	False	2024-02-01 19:48:27.000000 UTC	N/A	Disabled	
3228	660	svchost.exe	0xca82b7682080	3	-	0	False	2024-02-01 19:48:28.000000 UTC	N/A	Disabled	
4760	660	svchost.exe	0xca82b7757280	4	-	0	False	2024-02-01 19:48:28.000000 UTC	N/A	Disabled	
3132	660	svchost.exe	0xca82b7752280	9	-	0	False	2024-02-01 19:48:28.000000 UTC	N/A	Disabled	
5220	660	svchost.exe	0xca82b7792c0	5	-	1	False	2024-02-01 19:48:28.000000 UTC	N/A	Disabled	
5676	808	StartMenuXpex	0xca82b78c4080	12	-	1	False	2024-02-01 19:48:30.000000 UTC	N/A	Disabled	
5932	660	svchost.exe	0xca82b7b0a0c0	10	-	0	False	2024-02-01 19:48:30.000000 UTC	N/A	Disabled	
5948	808	RuntimeBroker.exe	0xca82b7a9b080	7	-	1	False	2024-02-01 19:48:30.000000 UTC	N/A	Disabled	
6128	808	SearchApp.exe	0xca82b7b10080	35	-	1	False	2024-02-01 19:48:30.000000 UTC	N/A	Disabled	
6216	808	RuntimeBroker.exe	0xca82b7b14240	15	-	1	False	2024-02-01 19:48:30.000000 UTC	N/A	Disabled	
6252	660	SearchIndexer.exe	0xca82b7daa0c0	16	-	0	False	2024-02-01 19:48:30.000000 UTC	N/A	Disabled	
6504	660	svchost.exe	0xca82b7b16240	4	-	0	False	2024-02-01 19:48:30.000000 UTC	N/A	Disabled	
6964	808	LockApp.exe	0xca82b7dc4080	12	-	1	False	2024-02-01 19:48:31.000000 UTC	N/A	Disabled	
7668	808	RuntimeBroker.exe	0xca82b81d42c0	7	-	1	False	2024-02-01 19:48:31.000000 UTC	N/A	Disabled	
5392	660	svchost.exe	0xca82b81ef2c0	4	-	0	False	2024-02-01 19:48:31.000000 UTC	N/A	Disabled	
2928	808	RuntimeBroker.exe	0xca82b82d80c0	6	-	1	False	2024-02-01 19:48:32.000000 UTC	N/A	Disabled	
7484	808	smartscreen.exe	0xca82b82d0340	17	-	1	False	2024-02-01 19:48:41.000000 UTC	N/A	Disabled	
7540	4568	SecurityHealth	0xca82b785b080	3	-	1	False	2024-02-01 19:48:41.000000 UTC	N/A	Disabled	
7580	660	SecurityHealth	0xca82b8586240	14	-	0	False	2024-02-01 19:48:41.000000 UTC	N/A	Disabled	
7684	4568	vmtoolsd.exe	0xca82b7d8e080	8	-	1	False	2024-02-01 19:48:41.000000 UTC	N/A	Disabled	
7780	4568	vmtoolsd.exe	0xca82b7d142c0	11	-	1	True	2024-02-01 19:48:42.000000 UTC	True	Disabled	
6984	808	wmiprvse.exe	0xca82b8046080	7	-	0	False	2024-02-01 19:48:46.000000 UTC	N/A	Disabled	
6500	660	svchost.exe	0xca82b84cc080	3	-	0	False	2024-02-01 19:48:49.000000 UTC	N/A	Disabled	
7272	2092	cmd.exe	0xca82b8490000	5	-	0	False	2024-02-01 19:48:49.000000 UTC	N/A	Disabled	
4628	4568	ChromeSetup.exe	0xca82b8310a300	4	-	1	True	2024-02-01 19:48:50.000000 UTC	N/A	Disabled	
7380	660	svchost.exe	0xca82b79841c0	5	-	0	False	2024-02-01 19:49:00.000000 UTC	N/A	Disabled	
4792	808	SkypeApp.exe	0xca82b84020c0	13	-	1	False	2024-02-01 19:49:06.000000 UTC	N/A	Disabled	
5644	4568	ChromeSetup.exe	0xca82b8310a300	4	-	1	True	2024-02-01 19:49:06.000000 UTC	N/A	Disabled	
5624	808	RuntimeBroker.exe	0xca82b8b182c0	1	-	1	False	2024-02-01 19:49:19.000000 UTC	N/A	Disabled	
7112	808	ApplicationFrameHost.exe	0xca82b8ba2080	3	-	1	False	2024-02-01 19:49:19.000000 UTC	N/A	Disabled	
5912	808	Webhost.exe	0xca82b8b60080	34	-	1	False	2024-02-01 19:49:19.000000 UTC	N/A	Disabled	
8420	808	dllhost.exe	0xca82b8e7b300	6	-	1	False	2024-02-01 19:49:21.000000 UTC	N/A	Disabled	
8728	808	vmtoolsd.exe	0xca82b7d142c0	11	-	1	False	2024-02-01 19:49:24.000000 UTC	N/A	Disabled	

What is the exact path of the executable for the malicious process?

1. Chạy để lọc kết quả cho "ChromeSetup".
➔ `vol3 -f “../memory.dmp” windows.filescan | grep "ChromeSetup"`
2. File thực thi nằm trong thư mục *Tải xuống* của nạn nhân. Nạn nhân đã tìm kiếm trình duyệt *Google Chrome*, gặp phải một liên kết độc hại và vô tình tải xuống cũng như thực thi phần mềm độc hại trên hệ thống của họ.

```

C:\WINDOWS\system32\cmd.exe /Q & powershell -c "cd C:\Users\voftha\Downloads\l59-Rammit\temp_extract_dir; .\thisIsTheMainTask_MING64.exe -~/Downloads/l59-Rammit/temp_extract_dir/main"; cd C:\Users\voftha\Downloads\l59-Rammit\temp_extract_dir; .\memory.dmp
2364 2876 WMIADAP.exe c:\Users\voftha\AppData\Local\Microsoft\Windows\CurrentVersion\Run~ 4 - 0 False 2024-02-01 19:52:25.000000 UTC N/A Disabled
6268 808 ShellExperienceHost.exe c:\Users\voftha\AppData\Local\Microsoft\Windows\CurrentVersion\Run~ 18 - 1 False 2024-02-01 19:52:26.000000 UTC N/A Disabled
2720 808 RuntimeBroker.exe c:\Users\voftha\AppData\Local\Microsoft\Windows\CurrentVersion\Run~ 6 - 1 False 2024-02-01 19:52:27.000000 UTC N/A Disabled
6096 6262 csrss.exe c:\Users\voftha\AppData\Local\Microsoft\Windows\CurrentVersion\Run~ 9 - 0 False 2024-02-01 19:53:28.000000 UTC N/A Disabled
3004 6252 SearchFilterHost.exe c:\Users\voftha\AppData\Local\Microsoft\Windows\CurrentVersion\Run~ 0 - 0 False 2024-02-01 19:53:28.000000 UTC N/A Disabled

.\thisIsTheMainTask_MING64.exe -~/Downloads/l59-Rammit/temp_extract_dir/main)
$ vol -f memory.dmp windows.filescan | grep "ChromeSetup"
usage: vol.exe [-h] [-c CONFIG] [--parallelism PLUGINS_THREADS] [-v] [-L LOG]
               [-o OUTPUT_DIR] [-q] [-f FILE] [--write-config]
               [--save-config SAVE_CONFIG] [--clear-cache]
               [--cache-path CACHE_PATH] [--offline [no URL]]
               [--filters FILTERS] [--hide-columns [HIDE_COLUMNS ...]]
               [--r RENDERER] [--single-location SINGLE_LOCATION]
               [--stackers [STACKERS ...]]
               [--single-swap-locations [SINGLE_SWAP_LOCATIONS ...]]
               --PLUGIN
vol.exe: error: File does not exist: C:\Users\voftha\Downloads\l59-Rammit\temp_extract_dir/memory.dmp

.\thisIsTheMainTask_MING64.exe -~/Downloads/l59-Rammit/temp_extract_dir/main)
$ vol -f ~/memory.dmp windows.filescan | grep "ChromeSetup"
c:\Users\voftha\Downloads\ChromeSetup.exe
c:\Users\voftha\Downloads\ChromeSetup.exe
c:\Users\voftha\Downloads\ChromeSetup.exe
Traceback (most recent call last):
  File "C:\Users\voftha\Downloads\chrome_ran.py", line 938, in _run_module_as_main
  File "C:\Users\voftha\Downloads\chrome_ran.py", line 88, in run_code
  File "C:\Users\voftha\AppData\Local\Programs\Python\Python312\Scripts\vol.exe\_main_.py", line 5, in <module>
  File "C:\Users\voftha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\cli\init_.py", line 934, in main
    CommandLine().run()
  File "C:\Users\voftha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\cli\init_.py", line 522, in run
    renderer.render(grid)
  File "C:\Users\voftha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\cli\text_renderer.py", line 329, in render
    grid.populate(visitor, outfd)
  File "C:\Users\voftha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\framework\renderers\_init_.py", line 318, in populate
    for level, item in self._generator:
      ^^^^^^^^^^^^^^
  File "C:\Users\voftha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\framework\plugins/windows/filescan.py", line 61, in _generator
    for fileobj in self.scan_files(fileobj.context, self.config['kernel']):
      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\voftha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\framework\plugins/windows/filescan.py", line 54, in scan_files
    result = poolscanner.PoolScanner.generate_pool_scan(
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\voftha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\framework\plugins/windows/poolscanner.py", line 503, in generate_pool_scan
    yield from cls.generate_pool_scan_extended(c)
  File "C:\Users\voftha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\framework\plugins/windows/poolscanner.py", line 431, in generate_pool_scan_extended
    for constraint, header in cls.pool_scan(
      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\voftha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\framework\plugins/windows/poolscanner.py", line 559, in pool_scan
    yield from layer.scan(context, scanner, progress_callback)
  File "C:\Users\voftha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\framework\interfaces\layers.py", line 258, in scan
    yield from scan_chunk(value)
           ^^^^^^^^^^^^^^^
  File "C:\Users\voftha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\framework\interfaces\layers.py", line 361, in _scan_chunk
    data = self.context.layers[layer.name].read(address, chunk_size)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\voftha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\framework\layers\linear.py", line 63, in read

```

Q3

Identifying network connections is crucial for understanding the malware's communication strategy. What IP address did the malware attempt to connect to?

1. Dùng *grep* một lần nữa, chúng tôi sẽ lọc kết quả để chỉ hiển thị những cái phù hợp với *PID* độc hại.
2. Dùng lệnh:
➔ `vol -f "../memory.dmp" windows.netscan | grep "ChromeSetup"`

```
File "C:\Users\votha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\framework\interfaces\layers.py",
line 258, in scan
yield from scan_chunk(value)
AAAAAAAAAAAAAAAAAAAA
File "C:\Users\votha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\framework\interfaces\layers.py",
line 361, in _scan_chunk
data += self._context.layers[layer_name].read(address, chunk_size)
AAAAAAAAAAAAAAAAAAAA
File "C:\Users\votha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\framework\layers\linear.py", line
63, in read
self._context.layers.read(layer, mapped_offset, mapped_length, pad)
File "C:\Users\votha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\framework\interfaces\layers.py",
line 636, in read
return self[layer].read(offset, length, pad)
AAAAAAAAAAAAAAAAAAAA
File "C:\Users\votha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\framework\layers\physical.py", li
ne 168, in read
data = self._file.read(length)
AAAAAAAAAAAAAAAAAAAA
File "C:\Users\votha\AppData\Local\Programs\Python\Python312\Lib\tempfile.py", line 499, in func_wrapper
return func(*args, **kwargs)
AAAAAAAAAAAAAAAAAAAA
KeyboardInterrupt

vvThais@ThaiHocHanh MINGW64 ~/Downloads/159-Ramnit/temp_extract_dir (main)
$ vol -f "../memory.dmp" windows.netscan | grep "ChromeSetup"
0xca82b7bebb40.0TCPv4 192.168.19.133an49682fin58.64.204.181 5202 CLOSED 4628 ChromeSetup.ex 2024-02-01 19:48:51.00
0000 UTC
0xca82b8bafb30 TCPv4 192.168.19.133 49682 58.64.204.181 5202 CLOSED 4628 ChromeSetup.ex 2024-02-01 19:48:51.00
0000 UTC
0xca82b8bc2b30 TCPv4 192.168.19.133 49682 58.64.204.181 5202 SYN_SENT 4628 ChromeSetup.ex 2024-02-01 19:
48:51.000000 UTC

vvThais@ThaiHocHanh MINGW64 ~/Downloads/159-Ramnit/temp_extract_dir (main)
$
```

Q4

To determine the specific geographical origin of the attack, Which city is associated with the IP address the malware communicated with?

1. Khi đã xác định được Ip của kẻ tấn công ta có thể dùng nhiều công cụ để kiểm tra xem kẻ tấn công đang sử dụng Ip ở đâu
2. Ở đây tôi sử dụng IPcheck.io để kiểm tra, cho thấy kẻ tấn công đang ở Hong Kong SAR China

IPCHECKER.IO

IP Reputation Checker Ping IP or Domain

Tiếng Việt (VI)

Dụng Cụ Kiểm Tra IP - Tra địa Chỉ IP

Hãy sử dụng công cụ check địa chỉ IP của chúng tôi để thực hiện việc tra cứu địa chỉ IP. Công cụ này sẽ tự động lấy địa chỉ IP hiện tại của bạn; ngoài ra, bạn cũng có thể nhập một địa chỉ IP hoặc tên miền bất kỳ. Công cụ sẽ tiến hành tra cứu và cung cấp các thông tin cần thiết, bao gồm vị trí địa lý của IP, chi tiết về mạng lưới, và nhiều thông tin khác.

Nhập Tên miền/IP để kiểm tra vị trí

58.64.204.181

Kiểm Tra IP

Criminal IP

IP: 58.64.204.181

Bang/Tỉnh: N/A

Vĩ độ: 22.3163

Mã quốc gia ISO: HK

ASN: 17444

Tên AS: HKBN Enterprise Solutions Limited

Tổ chức: HKBN Enterprise Solutions

ISP: HKBN Enterprise Solutions

Quốc gia: Hong Kong SAR China

Thành phố: Kowloon

Kinh độ: 114.1842

Mã Bưu Chính: N/A

Mức độ đe dọa: N/A

DB IP

IP: 58.64.204.181

Bang/Tỉnh: Kwai Tsing

Vĩ độ: 22.3702

Mã quốc gia ISO: HK

ASN: 17444

Tên AS: HKBNESL-AS-AP

Tổ chức: HKBN Enterprise Solutions Limited

ISP: HKBN Enterprise Solutions Limited

Quốc gia: Hong Kong

Thành phố: Kwai Chung

Kinh độ: 114.13

Mã Bưu Chính: N/A

Mức độ đe dọa: low

IP Location Services By:Criminal IP

Xem Trên Bản đồ

IP Location Services By:DB IP

Xem Trên Bản đồ

Q5

Hashes serve as unique identifiers for files, assisting in the detection of similar threats across different machines. What is the SHA1 hash of the malware executable?

1. Bước đầu tiên để lấy hash là trích xuất tệp thực thi từ bản dump bộ nhớ. Để làm điều này, chúng ta có thể tận dụng plugin của *Volatility* để trích xuất nội dung tệp từ hình ảnh. Sử dụng cú pháp bên dưới, chỉ định thư mục đầu ra cho bản dump và *PID* của tiến trình *ChromeSetup.exe* mà chúng ta tìm thấy trong
2. Sử dụng lệnh
➔ `vol -f "../memory.dmp" windows.dumpfile | grep "ChromeSetup"`

```
MINGW64 ~/Downloads/159-Ramnit/temp_extract_dir
file_handle = cls.dump_file_producer(
    AAAAAAAAAAAAAAAAAAAAAA
File "C:\Users\vottha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\framework\plugins\windows\dumpfil
es.py", line 108, in dump_file_producer
    filedata.write(data)
KeyboardInterrupt

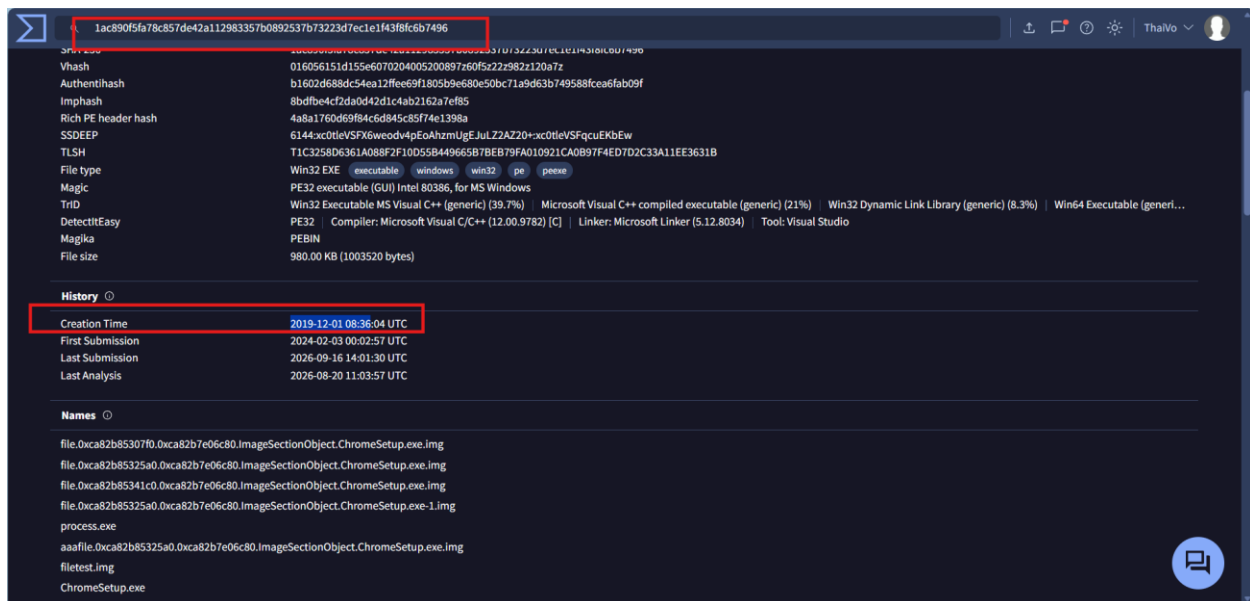
vvThais@ThaiHocHanh MINGW64 ~/Downloads/159-Ramnit/temp_extract_dir (main)
$ vol -f "../memory.dmp" windows.dumpfile --pid 4628
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
Cache FileObject FileName Result
ImageSectionObject 0xca82b8202cd0 winmm.dll file.0xca82b8202cd0.0xca82b79ab4b0.ImageSectionObject.winmm.dll.img
DataSectionObject 0xca82b85325a0 ChromeSetup.exe Error dumping file
ImageSectionObject 0xca82b85325a0 ChromeSetup.exe file.0xca82b85325a0.0xca82b7e06c80.ImageSectionObject.ChromeSetup.exe.
img
ImageSectionObject 0xca82b8215100 AcLayers.dll file.0xca82b8215100.0xca82b7ae1d60.ImageSectionObject.AcLayers.dll.img
ImageSectionObject 0xca82b8218300 sfc.dll file.0xca82b8218300.0xca82b8490050.ImageSectionObject.sfc.dll.img
ImageSectionObject 0xca82b821c950 winnrn.dll file.0xca82b821c950.0xca82b7de9c80.ImageSectionObject.winnrn.dll.img
ImageSectionObject 0xca82b8217810 msacm32.dll file.0xca82b8217810.0xca82b7dd6c80.ImageSectionObject.msacm32.dll.img
ImageSectionObject 0xca82b8536420 wsock32.dll file.0xca82b8536420.0xca82b7a69db0.ImageSectionObject.wsock32.dll.img
ImageSectionObject 0xca82b8218490 pnrpnsp.dll file.0xca82b8218490.0xca82b7e31ca0.ImageSectionObject.pnrpnsp.dll.img
ImageSectionObject 0xca82b82198e0 wshbth.dll file.0xca82b82198e0.0xca82b7bfac20.ImageSectionObject.wshbth.dll.img
ImageSectionObject 0xca82b821d8f0 nlaapi.dll file.0xca82b821d8f0.0xca82b7abbcc0.ImageSectionObject.nlaapi.dll.img
ImageSectionObject 0xca82b8539df0 NapiNSP.dll file.0xca82b8539df0.0xca82b7e4fc80.ImageSectionObject.NapiNSP.dll.img
ImageSectionObject 0xca82b8217cc0 winmmbase.dll file.0xca82b8217cc0.0xca82b7ae1b00.ImageSectionObject.winmmbase.dll.im
g
ImageSectionObject 0xca82b82187b0 sfc_os.dll file.0xca82b82187b0.0xca82b78b6a20.ImageSectionObject.sfc_os.dll.img
ImageSectionObject 0xca82b82179a0 winspool.drv file.0xca82b82179a0.0xca82b7e4cc80.ImageSectionObject.winspool.drv.img
ImageSectionObject 0xca82b8525a80 mssock.dll file.0xca82b8525a80.0xca82b7640cc0.ImageSectionObject.mssock.dll.img
ImageSectionObject 0xca82b8527380 apphelp.dll file.0xca82b8527380.0xca82b7105dc0.ImageSectionObject.apphelp.dll.img
ImageSectionObject 0xca82b82174f0 AcGenral.dll file.0xca82b82174f0.0xca82b7a8cb80.ImageSectionObject.AcGenral.dll.img
ImageSectionObject 0xca82b8523370 FWPUCLNT.DLL file.0xca82b8523370.0xca82b858ac80.ImageSectionObject.FWPUCLNT.DLL.img
ImageSectionObject 0xca82b8523820 rasadhlp.dll file.0xca82b8523820.0xca82b78cca20.ImageSectionObject.rasadhlp.dll.img
```

3. Và chỉ cần Sử dụng bấm lại details này nữa là xong

Q6

Examining the malware's development timeline can provide insights into its deployment. What is the compilation timestamp for the malware?

1. Khi đã Hash sh1 nó được thì copy đoạn mã đó sang VirusTotal
2. Kiểm tra phần details cho biết mọi thông tin về file trên



Q7

Identifying the domains associated with this malware is crucial for blocking future malicious communications and detecting any ongoing interactions with those domains within our network. Can you provide the domain connected to the malware?

1. Kiểm tra tại phần domain trong virustotal chỉ ra rất rõ Domain có sử dụng

